

Extraction Comparator Calculation Methodology

Input: manual triples are compared with LLM triples.

Manual triple shape: source -> [relation] -> target.

LLM row shape: source, source_type, relation, target, target_type.

Text is normalised: lowercase, remove punctuation, split into words.

Word similarity uses Jaccard = shared words / total unique words.

Source score compares manual source with LLM source.

Relation score compares manual relation with LLM relation.

Target score compares manual target with LLM target.

Triple score = source score x 0.4 + relation score x 0.2 + target score x 0.4.

For each manual row, the system chooses the LLM row with the highest triple score.

Threshold example: 60 percent means match cutoff is 0.60.

Partial cutoff = threshold x 0.5. With 60 percent threshold, partial cutoff is 30 percent.

If score >= threshold: Match.

If score >= partial cutoff and below threshold: Partial.

If score is below partial cutoff: Miss.

Partial matches receive half credit.

TP = matches + partials x 0.5.

FP = max of LLM count minus TP, or 0.

FN = misses + partials x 0.5.

Precision = TP / TP plus FP.

Recall = TP / TP plus FN.

F1 = 2 x precision x recall / precision plus recall.

Accuracy = TP / manual count.

Hallucinations = rounded FP.

Hallucination rate = FP / LLM count.

Best saved result ranks by F1, accuracy, precision, recall, then lower hallucination rate.

In simple words: it rewards correct matches, gives partial credit, and penalises extra LLM triples.